

Appendix B: Project Code

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    12:09:34 03/16/2011
6  -- Design Name:
7  -- Module Name:    mux41 - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22 library work;
23
24 entity mux41 is
25     Port ( c : in  STD_LOGIC_VECTOR (3 downto 0);
26           s : in  STD_LOGIC_VECTOR (1 downto 0);
27           z : out STD_LOGIC);
28 end mux41;
29
30 architecture Behavioral of mux41 is
31     component mux21
32     port(
33         a : in std_logic;
34         b : in std_logic;
35         s : in std_logic;
36         y : out std_logic
37     );
38 end component;
39
40     signal v : std_logic;
41     signal w : std_logic;
42
43     begin
44         U1 : mux21
45             port map (
46                 a => v, b => w, s => s(1), y => z
47             );
48         U2 : mux21
49             port map (
50                 a => c(0), b => c(1), s => s(0), y => v
51             );
52         U3 : mux21
53             port map (
54                 a => c(2), b => c(3), s => s(0), y => w
55             );
56     end Behavioral;
```

```
1
2  # PlanAhead Generated physical constraints
3
4  NET "c[0]" LOC = G18;
5  NET "c[1]" LOC = H18;
6  NET "c[2]" LOC = K18;
7  NET "c[3]" LOC = K17;
8  NET "s[0]" LOC = B18;
9  NET "s[1]" LOC = D18;
10 NET "z" LOC = J14;
11
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:   16:31:46 03/23/2011
6  -- Design Name:
7  -- Module Name:   F:/Senior Project/mux41/mux41_TB.vhd
8  -- Project Name:  mux41
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: mux41
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30  USE ieee.numeric_std.ALL;
31
32  ENTITY mux41_TB IS
33  END mux41_TB;
34
35  ARCHITECTURE behavior OF mux41_TB IS
36
37      -- Component Declaration for the Unit Under Test (UUT)
38
39      COMPONENT mux41
40      PORT(
41          c : IN  std_logic_vector(3 downto 0);
42          s : IN  std_logic_vector(1 downto 0);
43          z : OUT std_logic
44      );
45      END COMPONENT;
46
47
48      --Inputs
49      signal c : std_logic_vector(3 downto 0) := (others => '0');
50      signal s : std_logic_vector(1 downto 0) := (others => '0');
51
52      --Outputs
53      signal z : std_logic;
54      -- No clocks detected in port list. Replace <clock> below with
55      -- appropriate port name
56
```

```
57
58
59 BEGIN
60
61     -- Instantiate the Unit Under Test (UUT)
62     uut: mux41 PORT MAP (
63         c => c,
64         s => s,
65         z => z
66     );
67
68
69
70     -- Stimulus process
71     stim_proc: process
72     begin
73         c <= "0000";
74         s <= "00";
75
76         for i in 0 to 2 loop
77             for j in 0 to 4 loop
78                 wait for 10 ns;
79                 c <= std_logic_vector( unsigned(c) +1);
80             end loop;
81             s <= std_logic_vector( unsigned(s)+1);
82         end loop;
83
84         wait;
85     end process;
86
87 END;
88
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    12:48:58 04/13/2011
6  -- Design Name:
7  -- Module Name:    mux41b - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 -- Uncomment the following library declaration if using
24 -- arithmetic functions with Signed or Unsigned values
25 --use IEEE.NUMERIC_STD.ALL;
26
27 -- Uncomment the following library declaration if instantiating
28 -- any Xilinx primitives in this code.
29 --library UNISIM;
30 --use UNISIM.VComponents.all;
31
32 entity mux41b is
33     Port ( c : in  STD_LOGIC_VECTOR (3 downto 0);
34           s : in  STD_LOGIC_VECTOR (1 downto 0);
35           z : out STD_LOGIC);
36 end mux41b;
37
38 architecture Behavioral of mux41b is
39
40 begin
41
42     z <= (not s(1) and not s(0) and c(0))
43         or(not(s1) and s(0) and c(1))
44         or(s(1) and not s(0) and c(2))
45         or(s(1) and s(0) and c(3));
46
47
48 end Behavioral;
49
50
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    14:07:42 03/16/2011
6  -- Design Name:
7  -- Module Name:    mux41c - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19  -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity mux41c is
24     Port ( c : in  STD_LOGIC_VECTOR (3 downto 0);
25           s : in  STD_LOGIC_VECTOR (1 downto 0);
26           z : out STD_LOGIC);
27 end mux41c;
28
29 architecture Behavioral of mux41c is
30
31 begin
32     p1: process(c, s)
33     begin
34         case s is
35             when "00" => z <= c(0);
36             when "01" => z <= c(1);
37             when "10" => z <= c(2);
38             when "11" => z <= c(3);
39             when others => z <= c(0);
40         end case;
41     end process;
42 end Behavioral;
43
44
```

```
1
2  # PlanAhead Generated physical constraints
3
4  NET "c[0]" LOC = G18;
5  NET "c[1]" LOC = H18;
6  NET "c[2]" LOC = K18;
7  NET "c[3]" LOC = K17;
8  NET "s[0]" LOC = B18;
9  NET "s[1]" LOC = D18;
10 NET "z" LOC = J14;
11
```



```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:   14:36:30 03/16/2011
6  -- Design Name:
7  -- Module Name:   E:/Senior Project/mux41c/mux41c_TB.vhd
8  -- Project Name:  mux41c
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: mux41c
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30  USE ieee.numeric_std.ALL;
31
32
33  ENTITY mux41c_TB IS
34  END mux41c_TB;
35
36  ARCHITECTURE behavior OF mux41c_TB IS
37
38      -- Component Declaration for the Unit Under Test (UUT)
39
40      COMPONENT mux41c
41      PORT(
42          c : IN  std_logic_vector(3 downto 0);
43          s : IN  std_logic_vector(1 downto 0);
44          z : OUT std_logic
45      );
46      END COMPONENT;
47
48
49      --Inputs
50      signal c : std_logic_vector(3 downto 0) := (others => '0');
51      signal s : std_logic_vector(1 downto 0) := (others => '0');
52
53      --Outputs
54      signal z : std_logic;
55      -- No clocks detected in port list. Replace <clock> below with
56      -- appropriate port name
```

```
57
58
59
60 BEGIN
61
62     -- Instantiate the Unit Under Test (UUT)
63     uut: mux41c PORT MAP (
64         c => c,
65         s => s,
66         z => z
67     );
68
69
70
71     -- Stimulus process
72     stim_proc: process
73     begin
74         c <= "0000";
75         s <= "00";
76
77         for i in 0 to 2 loop
78             for j in 0 to 4 loop
79                 wait for 10 ns;
80                 c <= std_logic_vector( unsigned(c) +1);
81             end loop;
82             s <= std_logic_vector( unsigned(s)+1);
83         end loop;
84
85         wait;
86     end process;
87
88 END;
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    14:59:18 03/16/2011
6  -- Design Name:
7  -- Module Name:    mux44 - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19  -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23
24 entity mux44 is
25     Port ( x : in  STD_LOGIC_VECTOR (15 downto 0);
26           s : in  STD_LOGIC_VECTOR (1 downto 0);
27           z : out STD_LOGIC_VECTOR (3 downto 0));
28 end mux44;
29
30 architecture Behavioral of mux44 is
31
32
33 begin
34     p1: process(x, s)
35     begin
36         case s is
37             when "00" => z <= x(3 downto 0);
38             when "01" => z <= x(7 downto 4);
39             when "10" => z <= x(11 downto 8);
40             when "11" => z <= x(15 downto 12);
41             when others => z <= x(3 downto 0);
42         end case;
43     end process;
44 end Behavioral;
45
46
47
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:   10:41:10 04/05/2011
6  -- Design Name:
7  -- Module Name:   H:/Senior Project/mux44/mux44_TB.vhd
8  -- Project Name:  mux44
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: mux44
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30  USE ieee.numeric_std.ALL;
31
32  ENTITY mux44_TB IS
33  END mux44_TB;
34
35  ARCHITECTURE behavior OF mux44_TB IS
36
37      -- Component Declaration for the Unit Under Test (UUT)
38
39      COMPONENT mux44
40      PORT(
41          x : IN  std_logic_vector(15 downto 0);
42          s : IN  std_logic_vector(1 downto 0);
43          z : OUT std_logic_vector(3 downto 0)
44      );
45      END COMPONENT;
46
47
48      --Inputs
49      signal x : std_logic_vector(15 downto 0) := (others => '0');
50      signal s : std_logic_vector(1 downto 0) := (others => '0');
51
52      --Outputs
53      signal z : std_logic_vector(3 downto 0);
54      -- No clocks detected in port list. Replace <clock> below with
55      -- appropriate port name
56
```

```
57
58
59 BEGIN
60
61     -- Instantiate the Unit Under Test (UUT)
62     uut: mux44 PORT MAP (
63         x => x,
64         s => s,
65         z => z
66     );
67
68
69     -- Stimulus process
70     stim_proc: process
71     begin
72         x <= "000000000000000000";
73         s <= "00";
74
75         for i in 0 to 2 loop
76             for j in 0 to 16 loop
77                 wait for 10 ns;
78                 x <= std_logic_vector( unsigned(x) +1);
79             end loop;
80             s <= std_logic_vector( unsigned(s)+1);
81         end loop;
82
83         wait;
84     end process;
85
86 END;
87
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    15:15:56 03/16/2011
6  -- Design Name:
7  -- Module Name:    counter - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.all;
22 use IEEE.STD_LOGIC_unsigned.all;
23
24
25 entity counter is
26     generic(N : integer := 8);
27     Port (
28         clr : in  STD_LOGIC;
29         clk : in  STD_LOGIC;
30         q_counter : out  STD_LOGIC_VECTOR(N-1 downto 0)
31     );
32 end counter;
33
34 architecture Behavioral of counter is
35     signal count : STD_LOGIC_VECTOR(N-1 downto 0);
36
37 begin
38     process(clk, clr)
39     begin
40         if clr = '1' then
41             count <= (others => '0');
42         elsif clk'event and clk = '1' then
43             count <= count +1;
44         end if;
45     end process;
46
47     q_counter <= count;
48
49 end Behavioral;
50
51
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:   14:51:41 03/29/2011
6  -- Design Name:
7  -- Module Name:   E:/Senior Project/counter/counter_TB.vhd
8  -- Project Name:  counter
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: counter
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30
31  -- Uncomment the following library declaration if using
32  -- arithmetic functions with Signed or Unsigned values
33  --USE ieee.numeric_std.ALL;
34
35  ENTITY counter_TB IS
36  END counter_TB;
37
38  ARCHITECTURE behavior OF counter_TB IS
39
40      -- Component Declaration for the Unit Under Test (UUT)
41
42      COMPONENT counter
43      PORT(
44          clr : IN  std_logic;
45          clk : IN  std_logic;
46          q_counter : OUT  std_logic_vector(7 downto 0)
47      );
48      END COMPONENT;
49
50
51  --Inputs
52  signal clr : std_logic := '0';
53  signal clk : std_logic := '0';
54
55  --Outputs
56  signal q_counter : std_logic_vector(7 downto 0);
```

```
57
58     -- Clock period definitions
59     constant clk_period : time := 10 ns;
60
61 BEGIN
62
63     -- Instantiate the Unit Under Test (UUT)
64     uut: counter PORT MAP (
65         clr => clr,
66         clk => clk,
67         q_counter => q_counter
68     );
69
70     -- Clock process definitions
71     clk_process :process
72     begin
73         clk <= '0';
74         wait for clk_period/2;
75         clk <= '1';
76         wait for clk_period/2;
77     end process;
78
79
80     -- Stimulus process
81     stim_proc: process
82     begin
83         clr <= '1';
84         wait for 10 ns;
85         clr <= '0';
86         wait for 100 ns;
87     end process;
88
89 END;
90
```



```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    21:32:43 03/22/2011
6  -- Design Name:
7  -- Module Name:    clkdiv - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19  -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22 use IEEE.STD_LOGIC_unsigned.all;
23
24
25 entity clkdiv is
26     Port ( mclk : in  STD_LOGIC;
27           clr : in  STD_LOGIC;
28           clk190 : out STD_LOGIC;
29           clk48 : out STD_LOGIC
30
31         );
32 end clkdiv;
33
34 architecture Behavioral of clkdiv is
35
36
37     signal q : STD_LOGIC_VECTOR(23 downto 0);
38
39
40 begin
41
42     --clock divider
43     process(mclk, clr)
44     begin
45         if clr = '1' then
46             q <= X"000000";
47         elsif mclk'event and mclk = '1' then
48             q <= q+1;
49         end if;
50     end process;
51
52     clk48 <= q(20);    --48Hz
53     clk190 <= q(18);  --190 Hz
54
55
56 end Behavioral;
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:   14:57:22 03/29/2011
6  -- Design Name:
7  -- Module Name:   E:/Senior Project/clkdiv/clkdiv_TB.vhd
8  -- Project Name:  clkdiv
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: clkdiv
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30
31  ENTITY clkdiv_TB IS
32  END clkdiv_TB;
33
34  ARCHITECTURE behavior OF clkdiv_TB IS
35
36      -- Component Declaration for the Unit Under Test (UUT)
37
38      COMPONENT clkdiv
39      PORT(
40          mclk : IN  std_logic;
41          clr  : IN  std_logic;
42          clk190 : OUT std_logic;
43          clk48 : OUT std_logic
44      );
45      END COMPONENT;
46
47
48      --Inputs
49      signal mclk : std_logic := '0';
50      signal clr  : std_logic := '0';
51
52      --Outputs
53      signal clk190 : std_logic;
54      signal clk48  : std_logic;
55
56
```

```
57
58     -- Clock period definitions
59     constant mclk_period : time := 10 ns;
60
61
62 BEGIN
63
64     -- Instantiate the Unit Under Test (UUT)
65     uut: clkdiv PORT MAP (
66         mclk => mclk,
67         clr  => clr,
68         clk190 => clk190,
69         clk48  => clk48
70     );
71
72     -- Clock process definitions
73     mclk_process :process
74     begin
75         mclk <= '0';
76         wait for mclk_period/2;
77         mclk <= '1';
78         wait for mclk_period/2;
79     end process;
80
81
82
83     -- Stimulus process
84     stim_proc: process
85     begin
86         -- hold reset state for 100 ns.
87         clr <= '1';
88         wait for 10 ns;
89         clr <= '0';
90         wait for 2000 ns;
91         wait;
92     end process;
93
94 END;
95
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:      21:40:35 03/22/2011
6  -- Design Name:
7  -- Module Name:      count8_top - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19  -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22 library work;
23
24 entity count8_top is
25     Port ( mclk : in  STD_LOGIC;
26           btn : in  STD_LOGIC_VECTOR (3 downto 3);
27           ld : out  STD_LOGIC_VECTOR (7 downto 0));
28 end count8_top;
29
30 architecture Behavioral of count8_top is
31
32     component clkdiv
33     port(
34         clr : in std_logic;
35         mclk : in std_logic;
36         clk3 : out std_logic
37     );
38 end component;
39
40     component counter
41     generic(N : integer := 8);
42     Port (
43         clr : in  STD_LOGIC;
44         clk : in  STD_LOGIC;
45         q_counter : out  STD_LOGIC_VECTOR(N-1 downto 0)
46     );
47 end component;
48
49     signal clk3 : std_logic;
50
51     begin
52     U1 : clkdiv
53         port map (
54             clk3 => clk3, clr => btn(3), mclk => mclk);
55
56     U2: counter
```

```
57     generic map(  
58         N => 8)  
59     port map(  
60         clk => clk3, clr => btn(3), q_counter => ld(7 downto 0));  
61  
62     end Behavioral;  
63  
64
```

```
1
2 # PlanAhead Generated physical constraints
3 NET "mclk" LOC = B8;
4 NET "btn[3]" LOC = H13;
5 NET "ld[0]" LOC = J14;
6 NET "ld[1]" LOC = J15;
7 NET "ld[2]" LOC = K15;
8 NET "ld[3]" LOC = K14;
9 NET "ld[4]" LOC = E17;
10 NET "ld[5]" LOC = P1;
11 NET "ld[6]" LOC = F4;
12 NET "ld[7]" LOC = R4;
13
14
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:   14:30:28 03/29/2011
6  -- Design Name:
7  -- Module Name:   E:/Senior Project/count8_top/count8_top_TB.vhd
8  -- Project Name:  count8_top
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: count8_top
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27 -----
28 LIBRARY ieee;
29 USE ieee.std_logic_1164.ALL;
30
31 -- Uncomment the following library declaration if using
32 -- arithmetic functions with Signed or Unsigned values
33 --USE ieee.numeric_std.ALL;
34
35 ENTITY count8_top_TB IS
36 END count8_top_TB;
37
38 ARCHITECTURE behavior OF count8_top_TB IS
39
40     -- Component Declaration for the Unit Under Test (UUT)
41
42     COMPONENT count8_top
43     PORT(
44         mclk : IN  std_logic;
45         btn  : IN  std_logic_vector(3 downto 3);
46         ld   : OUT std_logic_vector(7 downto 0)
47     );
48     END COMPONENT;
49
50
51     --Inputs
52     signal mclk : std_logic := '0';
53     signal btn  : std_logic_vector(3 downto 3) := (others => '0');
54
55     --Outputs
56     signal ld : std_logic_vector(7 downto 0);
```

```
57
58     -- Clock period definitions
59     constant mclk_period : time := 10 ns;
60
61 BEGIN
62
63     -- Instantiate the Unit Under Test (UUT)
64     uut: count8_top PORT MAP (
65         mclk => mclk,
66         btn => btn,
67         ld => ld
68     );
69
70     -- Clock process definitions
71     mclk_process :process
72     begin
73         mclk <= '0';
74         wait for mclk_period/2;
75         mclk <= '1';
76         wait for mclk_period/2;
77     end process;
78
79
80     -- Stimulus process
81     stim_proc: process
82     begin
83         -- hold reset state for 100 ns.
84         btn(3) <= '1';
85         wait for 10 ns;
86         btn(3) <= '0';
87         wait for 1000 ns;
88         wait;
89     end process;
90
91 END;
92
```



```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    15:40:40 03/16/2011
6  -- Design Name:
7  -- Module Name:    hex7seg_le - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity hex7seg_le is
24     Port ( x : in  STD_LOGIC_VECTOR (3 downto 0);
25           a_to_g : out STD_LOGIC_VECTOR (6 downto 0));
26 end hex7seg_le;
27
28 architecture Behavioral of hex7seg_le is
29
30 begin
31     a_to_g(6)  <= (not x(3) and not x(2) and not x(1) and x(0)) --a
32                or (not x(3) and  x(2) and not x(1) and not x(0))
33                or (x(3) and x(2) and not x(1) and  x(0))
34                or (x(3) and not x(2) and x(1) and x(0));
35     a_to_g(5)  <= (x(2) and x(1) and not x(0)) --b
36                or (x(3) and x(1) and x(0))
37                or (not x(3) and x(2) and not x(1) and x(0))
38                or (x(3) and x(2) and not x(1) and not x(0));
39     a_to_g(4)  <= (not x(3) and not x(2) and x(1) and not x(0)) --c
40                or (x(3) and  x(2) and x(1))
41                or (x(3) and x(2) and not x(0));
42     a_to_g(3)  <= (not x(3) and not x(2) and not x(1) and x(0)) --d
43                or (not x(3) and  x(2) and not x(1) and not x(0))
44                or (x(3) and not x(2) and x(1) and not x(0))
45                or (x(2) and x(1) and x(0));
46     a_to_g(2)  <= (not x(3) and x(0)) --e
47                or (not x(3) and  x(2) and not x(1))
48                or (not x(2) and not x(1) and x(0));
49     a_to_g(1)  <= (not x(3) and not x(2) and x(0)) --f
50                or (not x(3) and not x(2) and x(1))
51                or (not x(3) and x(1) and x(0))
52                or (x(3) and x(2) and not x(1) and x(0));
53     a_to_g(0)  <= (not x(3) and not x(2) and not x(1)) --g
54                or (x(3) and x(2) and not x(1) and not x(0))
55                or (not x(3) and x(2) and x(1) and x(0));
56
```

```
57   end Behavioral;  
58  
59
```

```
1
2  # PlanAhead Generated physical constraints
3
4  NET "a_to_g[0]" LOC = H14;
5  NET "a_to_g[1]" LOC = J17;
6  NET "a_to_g[2]" LOC = G14;
7  NET "a_to_g[3]" LOC = D16;
8  NET "a_to_g[4]" LOC = D17;
9  NET "a_to_g[5]" LOC = F18;
10 NET "a_to_g[6]" LOC = L18;
11 NET "x[0]" LOC = G18;
12 NET "x[1]" LOC = H18;
13 NET "x[2]" LOC = K18;
14 NET "x[3]" LOC = K17;
15
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:   16:04:01 03/16/2011
6  -- Design Name:
7  -- Module Name:   E:/misc/COIN330/hex7seg_le/hex7seg_le_TB.vhd
8  -- Project Name:  hex7seg_le
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: hex7seg_le
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30  USE ieee.numeric_std.ALL;
31
32  ENTITY hex7seg_le_TB IS
33  END hex7seg_le_TB;
34
35  ARCHITECTURE behavior OF hex7seg_le_TB IS
36
37      -- Component Declaration for the Unit Under Test (UUT)
38
39      COMPONENT hex7seg_le
40      PORT(
41          x : IN  std_logic_vector(3 downto 0);
42          a_to_g : OUT  std_logic_vector(6 downto 0)
43      );
44      END COMPONENT;
45
46
47      --Inputs
48      signal x : std_logic_vector(3 downto 0) := (others => '0');
49
50      --Outputs
51      signal a_to_g : std_logic_vector(6 downto 0);
52      -- No clocks detected in port list. Replace <clock> below with
53      -- appropriate port name
54
55
56
```

```
57 BEGIN
58
59 -- Instantiate the Unit Under Test (UUT)
60 uut: hex7seg_le PORT MAP (
61     x => x,
62     a_to_g => a_to_g
63 );
64
65 -- Stimulus process
66 stim_proc: process
67 begin
68
69     x <= "0000";
70     for i in 0 to 16 loop
71         wait for 25 ns;
72         x <= std_logic_vector(unsigned(x) +1);
73     end loop;
74
75     wait;
76 end process;
77
78 END;
79
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    16:13:51 03/16/2011
6  -- Design Name:
7  -- Module Name:    hex7seg - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23
24
25 entity hex7seg is
26     Port ( x : in  STD_LOGIC_VECTOR (3 downto 0);
27           a_to_g : out STD_LOGIC_VECTOR (6 downto 0));
28 end hex7seg;
29
30 architecture Behavioral of hex7seg is
31
32 begin
33 process(x)
34 begin
35 case x is
36     when X"0" => a_to_g <= "0000001"; --0
37     when X"1" => a_to_g <= "1001111"; --1
38     when X"2" => a_to_g <= "0010010"; --2
39     when X"3" => a_to_g <= "0000110"; --3
40     when X"4" => a_to_g <= "1001100"; --4
41     when X"5" => a_to_g <= "0100100"; --5
42     when X"6" => a_to_g <= "0100000"; --6
43     when X"7" => a_to_g <= "0001101"; --7
44     when X"8" => a_to_g <= "0000000"; --8
45     when X"9" => a_to_g <= "0000100"; --9
46     when X"A" => a_to_g <= "0001000"; --A
47     when X"B" => a_to_g <= "1100000"; --B
48     when X"C" => a_to_g <= "0110001"; --C
49     when X"D" => a_to_g <= "1000010"; --D
50     when X"E" => a_to_g <= "0110000"; --E
51     when others => a_to_g <= "0111000"; --F
52 end case;
53 end process;
54 end Behavioral;
55
56
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    21:34:11 04/06/2011
6  -- Design Name:
7  -- Module Name:    F:/Senior Project/hex7seg/hex7seg_TB.vhd
8  -- Project Name:   hex7seg
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: hex7seg
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30  USE ieee.numeric_std.ALL;
31
32  ENTITY hex7seg_TB IS
33  END hex7seg_TB;
34
35  ARCHITECTURE behavior OF hex7seg_TB IS
36
37      -- Component Declaration for the Unit Under Test (UUT)
38
39      COMPONENT hex7seg
40      PORT(
41          x : IN  std_logic_vector(3 downto 0);
42          a_to_g : OUT std_logic_vector(6 downto 0)
43      );
44      END COMPONENT;
45
46
47      --Inputs
48      signal x : std_logic_vector(3 downto 0) := (others => '0');
49
50      --Outputs
51      signal a_to_g : std_logic_vector(6 downto 0);
52      -- No clocks detected in port list. Replace <clock> below with
53      -- appropriate port name
54
55
56  BEGIN
```

```
57
58     -- Instantiate the Unit Under Test (UUT)
59     uut: hex7seg PORT MAP (
60         x => x,
61         a_to_g => a_to_g
62     );
63
64
65     -- Stimulus process
66     stim_proc: process
67     begin
68         x <= "0000";
69         for i in 0 to 16 loop
70             wait for 25 ns;
71             x <= std_logic_vector(unsigned(x) +1);
72         end loop;
73         wait;
74     end process;
75
76     END;
77
```



```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    19:27:01 03/22/2011
6  -- Design Name:
7  -- Module Name:    hex7seg_top - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19  -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23
24 entity hex7seg_top is
25     Port ( sw : in  STD_LOGIC_VECTOR (3 downto 0);
26           a_to_g : out STD_LOGIC_VECTOR (6 downto 0);
27           an : out  STD_LOGIC_VECTOR (3 downto 0);
28           dp : out  STD_LOGIC);
29 end hex7seg_top;
30
31 architecture Behavioral of hex7seg_top is
32 component hex7seg is
33     port(
34         x : in STD_LOGIC_VECTOR(3 downto 0);
35         a_to_g : out STD_LOGIC_VECTOR(6 downto 0)
36     );
37 end component;
38
39 begin
40 an <= "0000"; --all digits on
41 dp <= '1'; --dp off
42
43 D4: hex7seg port map
44     (x => sw,
45      a_to_g => a_to_g
46     );
47
48 end Behavioral;
49
50
```

```
1
2 # PlanAhead Generated physical constraints
3 NET "a_to_g[0]" LOC = H14;
4 NET "a_to_g[1]" LOC = J17;
5 NET "a_to_g[2]" LOC = G14;
6 NET "a_to_g[3]" LOC = D16;
7 NET "a_to_g[4]" LOC = D17;
8 NET "a_to_g[5]" LOC = F18;
9 NET "a_to_g[6]" LOC = L18;
10 NET "an[0]" LOC = F17;
11 NET "an[1]" LOC = H17;
12 NET "an[2]" LOC = C18;
13 NET "an[3]" LOC = F15;
14 NET "dp" LOC = C17;
15 NET "sw[0]" LOC = G18;
16 NET "sw[1]" LOC = H18;
17 NET "sw[2]" LOC = K18;
18 NET "sw[3]" LOC = K17;
19
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    21:51:57 04/06/2011
6  -- Design Name:
7  -- Module Name:    F:/Senior Project/hex7seg_top/hex7seg_top_TB.vhd
8  -- Project Name:   hex7seg_top
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: hex7seg_top
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30  USE ieee.numeric_std.ALL;
31
32  ENTITY hex7seg_top_TB IS
33  END hex7seg_top_TB;
34
35  ARCHITECTURE behavior OF hex7seg_top_TB IS
36
37      -- Component Declaration for the Unit Under Test (UUT)
38
39      COMPONENT hex7seg_top
40      PORT(
41          sw : IN  std_logic_vector(3 downto 0);
42          a_to_g : OUT std_logic_vector(6 downto 0);
43          an : OUT  std_logic_vector(3 downto 0);
44          dp : OUT  std_logic
45      );
46      END COMPONENT;
47
48
49      --Inputs
50      signal sw : std_logic_vector(3 downto 0) := (others => '0');
51
52      --Outputs
53      signal a_to_g : std_logic_vector(6 downto 0);
54      signal an : std_logic_vector(3 downto 0);
55      signal dp : std_logic;
56      -- No clocks detected in port list. Replace <clock> below with
```

```
57      -- appropriate port name
58
59
60  BEGIN
61
62      -- Instantiate the Unit Under Test (UUT)
63      uut: hex7seg_top PORT MAP (
64          sw => sw,
65          a_to_g => a_to_g,
66          an => an,
67          dp => dp
68      );
69
70
71
72      -- Stimulus process
73      stim_proc: process
74      begin
75          sw <= "0000";
76          for i in 0 to 16 loop
77              wait for 25 ns;
78              sw <= std_logic_vector(unsigned(sw) +1);
79          end loop;
80          wait;
81      end process;
82
83  END;
84
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    13:08:40 04/13/2011
6  -- Design Name:
7  -- Module Name:    mux7seg - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity mux7seg is
24     Port ( btn : in  STD_LOGIC_VECTOR (3 downto 0);
25           a_to_g : out STD_LOGIC_VECTOR (6 downto 0);
26           an : out  STD_LOGIC_VECTOR (3 downto 0));
27 end mux7seg;
28
29 architecture Behavioral of mux7seg is
30
31     component hex7seg is
32         Port ( x : in  STD_LOGIC_VECTOR (3 downto 0);
33               a_to_g : out STD_LOGIC_VECTOR (6 downto 0));
34     end component;
35
36     component mux44 is
37         Port ( x : in  STD_LOGIC_VECTOR (15 downto 0);
38               s : in  STD_LOGIC_VECTOR (1 downto 0);
39               z : out  STD_LOGIC_VECTOR (3 downto 0));
40     end component;
41
42     signal digit : STD_LOGIC_VECTOR (3 downto 0);
43     signal s : STD_LOGIC_VECTOR (1 downto 0);
44     signal x : STD_LOGIC_VECTOR (15 downto 0);
45
46     begin
47         x <= X"1234";
48
49         U1 : hex7seg port map( a_to_g => a_to_g, x => digit);
50
51         U2 : mux44 port map( s=> s, x => x, z => digit);
52
53         s(0) <= btn(3) or btn(1);
54         s(1) <= btn(3) or btn(2);
55         an(1) <= not(btn(1));
56         an(0) <= not(btn(0));
```

```
57  an(2) <= not(btn(2));  
58  an(3) <= not(btn(3));  
59  end Behavioral;  
60  
61
```

```
1
2
3  NET "btn[0]" LOC = B18;
4  NET "btn[1]" LOC = D18;
5  NET "btn[3]" LOC = H13;
6  NET "btn[2]" LOC = E18;
7  NET "a_to_g[0]" LOC = H14;
8  NET "a_to_g[1]" LOC = J17;
9  NET "a_to_g[2]" LOC = G14;
10 NET "a_to_g[3]" LOC = D16;
11 NET "a_to_g[4]" LOC = D17;
12 NET "a_to_g[5]" LOC = F18;
13 NET "a_to_g[6]" LOC = L18;
14 NET "an[0]" LOC = F17;
15 NET "an[1]" LOC = H17;
16 NET "an[2]" LOC = C18;
17 NET "an[3]" LOC = F15;
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    21:05:04 03/22/2011
6  -- Design Name:
7  -- Module Name:    x7seg - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity x7seg is
24     Port ( cclk : in  STD_LOGIC;
25           clr : in  STD_LOGIC;
26           x : in  STD_LOGIC_VECTOR (15 downto 0);
27           a_to_g : out STD_LOGIC_VECTOR (6 downto 0);
28           an : out  STD_LOGIC_VECTOR (3 downto 0));
29 end x7seg;
30
31 architecture Behavioral of x7seg is
32     component counter is
33         generic(N : integer := 2);
34         Port (
35             clr : in  STD_LOGIC;
36             clk : in  STD_LOGIC;
37             q_counter : out STD_LOGIC_VECTOR(N-1 downto 0)
38         );
39     end component;
40
41     component hex7seg
42     port(
43         x : in STD_LOGIC_VECTOR (3 downto 0);
44         a_to_g : out STD_LOGIC_VECTOR (6 downto 0)
45     );
46     end component;
47     component mux44
48     port(
49         s : in STD_LOGIC_VECTOR (1 downto 0);
50         x : in STD_LOGIC_VECTOR (15 downto 0);
51         z : out STD_LOGIC_VECTOR (3 downto 0)
52     );
53     end component;
54
55     signal nq0 : STD_LOGIC;
56     signal nq1 : STD_LOGIC;
```



```
57  signal digit : STD_LOGIC_VECTOR (3 downto 0);
58  signal q : STD_LOGIC_VECTOR (1 downto 0);
59
60  begin
61
62    U1 : hex7seg
63      port map(
64        a_to_g => a_to_g,
65        x => digit
66      );
67
68    nq1 <= not(q(1));
69    nq0 <= not(q(0));
70
71    U2 : mux44
72      port map(
73        s(0) => q(0),
74        s(1) => q(1),
75        x => x,
76        z => digit
77      );
78    U3 : counter
79      port map(
80        clk => cclk,
81        clr => clr,
82        q_counter => q(1 downto 0)
83      );
84
85    an(0) <= q(0) or q(1);
86    an(1) <= nq0 or q(1);
87    an(2) <= q(0) or nq1;
88    an(3) <= nq0 or nq1;
89
90
91  end Behavioral;
92
93
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:   15:43:06 04/13/2011
6  -- Design Name:
7  -- Module Name:   H:/Senior Project/Example 10/x7seg/xyseg_TB.vhd
8  -- Project Name:  x7seg
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: x7seg
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30
31
32  ENTITY xyseg_TB IS
33  END xyseg_TB;
34
35  ARCHITECTURE behavior OF xyseg_TB IS
36
37      -- Component Declaration for the Unit Under Test (UUT)
38
39      COMPONENT x7seg
40      PORT(
41          cclk : IN  std_logic;
42          clr  : IN  std_logic;
43          x    : IN  std_logic_vector(15 downto 0);
44          a_to_g : OUT std_logic_vector(6 downto 0);
45          an    : OUT std_logic_vector(3 downto 0)
46      );
47      END COMPONENT;
48
49
50      --Inputs
51      signal cclk : std_logic := '0';
52      signal clr  : std_logic := '0';
53      signal x    : std_logic_vector(15 downto 0) := (others => '0');
54
55      --Outputs
56      signal a_to_g : std_logic_vector(6 downto 0);
```

```
57     signal an : std_logic_vector(3 downto 0);
58
59     -- Clock period definitions
60     constant cclk_period : time := 10 ns;
61
62 BEGIN
63
64     -- Instantiate the Unit Under Test (UUT)
65     uut: x7seg PORT MAP (
66         cclk => cclk,
67         clr => clr,
68         x => x,
69         a_to_g => a_to_g,
70         an => an
71     );
72
73     -- Clock process definitions
74     cclk_process : process
75     begin
76         cclk <= '0';
77         wait for cclk_period/2;
78         cclk <= '1';
79         wait for cclk_period/2;
80     end process;
81
82
83     -- Stimulus process
84     stim_proc: process
85     begin
86
87         x <= X"1234";
88         clr <= '1';
89         wait for 10 ns;
90         clr <= '0';
91         wait for 1000 ns;
92         wait;
93     end process;
94
95 END;
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    22:59:50 03/22/2011
6  -- Design Name:
7  -- Module Name:    x7seg_top - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23
24
25 entity x7seg_top is
26     Port ( mclk : in  STD_LOGIC;
27           btn : in  STD_LOGIC_VECTOR (3 downto 3);
28           a_to_g : out STD_LOGIC_VECTOR (6 downto 0);
29           an : out  STD_LOGIC_VECTOR (3 downto 0);
30           dp : out  STD_LOGIC);
31 end x7seg_top;
32
33 architecture Behavioral of x7seg_top is
34 component x7seg is
35     Port (
36         cclk : in  STD_LOGIC;
37         clr : in  STD_LOGIC;
38         x : in  STD_LOGIC_VECTOR (15 downto 0);
39         a_to_g : out STD_LOGIC_VECTOR (6 downto 0);
40         an : out  STD_LOGIC_VECTOR (3 downto 0)
41     );
42 end component;
43
44 component clkdiv
45     port(
46         clr : in STD_LOGIC;
47         mclk : in STD_LOGIC;
48         clk190 : out STD_LOGIC
49     );
50 end component;
51
52 signal x : STD_LOGIC_VECTOR(15 downto 0);
53 signal clk190: STD_LOGIC;
54
55 begin
56
```

```
57     x <= X"1234";
58
59     U1 : clkdiv port map
60         (clr => btn(3), mclk => mclk, clk190 =>clk190);
61
62     U2 : x7seg port map
63         (x => x, cclk => clk190, clr => btn(3), a_to_g => a_to_g,
64          an => an);
65     dp <= '1';
66
67 end Behavioral;
68
69
```

```
1
2 # PlanAhead Generated physical constraints
3 NET "a_to_g[0]" LOC = H14;
4 NET "a_to_g[1]" LOC = J17;
5 NET "a_to_g[2]" LOC = G14;
6 NET "a_to_g[3]" LOC = D16;
7 NET "a_to_g[4]" LOC = D17;
8 NET "a_to_g[5]" LOC = F18;
9 NET "a_to_g[6]" LOC = L18;
10 NET "an[0]" LOC = F17;
11 NET "an[1]" LOC = H17;
12 NET "an[2]" LOC = C18;
13 NET "an[3]" LOC = F15;
14 NET "btn[3]" LOC = H13;
15 NET "dp" LOC = C17;
16 NET "mclk" LOC = B8;
17
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    15:39:14 04/13/2011
6  -- Design Name:
7  -- Module Name:    H:/Senior Project/Example 10/x7seg_top/x7seg_top_TB.vhd
8  -- Project Name:   x7seg_top
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: x7seg_top
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30  USE ieee.numeric_std.ALL;
31
32  ENTITY x7seg_top_TB IS
33  END x7seg_top_TB;
34
35  ARCHITECTURE behavior OF x7seg_top_TB IS
36
37      -- Component Declaration for the Unit Under Test (UUT)
38
39      COMPONENT x7seg_top
40      PORT(
41          mclk : IN  std_logic;
42          btn  : IN  std_logic_vector(3 downto 3);
43          a_to_g : OUT std_logic_vector(6 downto 0);
44          an   : OUT std_logic_vector(3 downto 0);
45          dp   : OUT std_logic
46      );
47      END COMPONENT;
48
49
50      --Inputs
51      signal mclk : std_logic := '0';
52      signal btn  : std_logic_vector(3 downto 3) := (others => '0');
53
54      --Outputs
55      signal a_to_g : std_logic_vector(6 downto 0);
56      signal an   : std_logic_vector(3 downto 0);
```

```
57     signal dp : std_logic;
58
59     -- Clock period definitions
60     constant mclk_period : time := 10 ns;
61
62 BEGIN
63
64     -- Instantiate the Unit Under Test (UUT)
65     uut: x7seg_top PORT MAP (
66         mclk => mclk,
67         btn  => btn,
68         a_to_g => a_to_g,
69         an  => an,
70         dp  => dp
71     );
72
73     -- Clock process definitions
74     mclk_process : process
75     begin
76         mclk <= '0';
77         wait for mclk_period/2;
78         mclk <= '1';
79         wait for mclk_period/2;
80     end process;
81
82
83     -- Stimulus process
84     stim_proc: process
85     begin
86         btn(3) <= '1';
87         wait for 10 ns;
88         btn(3) <= '0';
89         wait for 100 ns;
90
91         wait;
92     end process;
93
94 END;
```



```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    23:13:12 03/22/2011
6  -- Design Name:
7  -- Module Name:    x7segb - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19  -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22 use IEEE.STD_LOGIC_UNSIGNED.ALL;
23
24 entity x7segb is
25     Port ( x : in  STD_LOGIC_VECTOR (15 downto 0);
26           clk : in  STD_LOGIC;
27           clr : in  STD_LOGIC;
28           a_to_g : out STD_LOGIC_VECTOR (6 downto 0);
29           an : out  STD_LOGIC_VECTOR (3 downto 0);
30           dp : out  STD_LOGIC);
31 end x7segb;
32
33 architecture Behavioral of x7segb is
34
35     signal s : STD_LOGIC_VECTOR(1 downto 0);
36     signal digit : STD_LOGIC_VECTOR(3 downto 0);
37     signal aen : STD_LOGIC_VECTOR(3 downto 0);
38     signal clkdiv : STD_LOGIC_VECTOR(20 downto 0);
39
40     begin
41
42         s <= clkdiv(20 downto 19);
43         dp <= '1';
44
45         --set aen(3 downto 0) for leading blanks
46         aen(3) <= x(15) or x(14) or x(13) or x(12);
47         aen(2) <= x(15) or x(14) or x(13) or x(12)
48             or x(11) or x(10) or x(9) or x(8);
49         aen(1) <= x(15) or x(14) or x(13) or x(12)
50             or x(11) or x(10) or x(9) or x(8)
51             or x(7) or x(6) or x(5) or x(4);
52         aen(0) <= '1'; --digit 0 always on
53
54         --Quad 4-to-1 MUX: mux44
55         process(s, x)
56             begin
```

```
57     case s is
58         when "00" => digit <= x(3 downto 0);
59         when "01" => digit <= x(7 downto 4);
60         when "10" => digit <= x(11 downto 8);
61         when others => digit <= x(15 downto 12);
62     end case;
63 end process;
64
65 --7 segment decoder: hex7seg
66 process(digit)
67 begin
68     case digit is
69         when X"0" => a_to_g <= "0000001"; --0
70         when X"1" => a_to_g <= "1001111"; --1
71         when X"2" => a_to_g <= "0010010"; --2
72         when X"3" => a_to_g <= "0000110"; --3
73         when X"4" => a_to_g <= "1001100"; --4
74         when X"5" => a_to_g <= "0100100"; --5
75         when X"6" => a_to_g <= "0100000"; --6
76         when X"7" => a_to_g <= "0001101"; --7
77         when X"8" => a_to_g <= "0000000"; --8
78         when X"9" => a_to_g <= "0000100"; --9
79         when X"A" => a_to_g <= "0001000"; --A
80         when X"B" => a_to_g <= "1100000"; --B
81         when X"C" => a_to_g <= "0110001"; --C
82         when X"D" => a_to_g <= "1000010"; --D
83         when X"E" => a_to_g <= "0110000"; --E
84         when others => a_to_g <= "0111000"; --F
85     end case;
86 end process;
87
88 --Digit select : ancode
89 process(s, aen)
90 begin
91     an <= "1111";
92     if aen(conv_integer(s)) = '1' then
93         an(conv_integer(s)) <= '0';
94     end if;
95 end process;
96
97 --Clock divider
98 process(clk, clr)
99 begin
100     if clr = '1' then
101         clkdiv <= (others => '0');
102     elsif clk'event and clk = '1' then
103         clkdiv <= clkdiv + 1;
104     end if;
105 end process;
106 end Behavioral;
107
108
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    17:45:21 04/08/2011
6  -- Design Name:
7  -- Module Name:    F:/Senior Project/x7segb/x7segb_TB.vhd
8  -- Project Name:   x7segb
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: x7segb
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30  USE ieee.numeric_std.ALL;
31
32  ENTITY x7segb_TB IS
33  END x7segb_TB;
34
35  ARCHITECTURE behavior OF x7segb_TB IS
36
37      -- Component Declaration for the Unit Under Test (UUT)
38
39      COMPONENT x7segb
40      PORT(
41          x : IN  std_logic_vector(15 downto 0);
42          clk : IN  std_logic;
43          clr : IN  std_logic;
44          a_to_g : OUT std_logic_vector(6 downto 0);
45          an : OUT  std_logic_vector(3 downto 0);
46          dp : OUT  std_logic
47      );
48      END COMPONENT;
49
50
51      --Inputs
52      signal x : std_logic_vector(15 downto 0) := (others => '0');
53      signal clk : std_logic := '0';
54      signal clr : std_logic := '0';
55
56      --Outputs
```

```
57     signal a_to_g : std_logic_vector(6 downto 0);
58     signal an : std_logic_vector(3 downto 0);
59     signal dp : std_logic;
60
61     -- Clock period definitions
62     constant clk_period : time := 10 ns;
63
64 BEGIN
65
66     -- Instantiate the Unit Under Test (UUT)
67     uut: x7segb PORT MAP (
68         x => x,
69         clk => clk,
70         clr => clr,
71         a_to_g => a_to_g,
72         an => an,
73         dp => dp
74     );
75
76     -- Clock process definitions
77     clk_process :process
78     begin
79         clk <= '0';
80         wait for clk_period/2;
81         clk <= '1';
82         wait for clk_period/2;
83     end process;
84
85     -- Stimulus process
86     stim_proc: process
87     begin
88         x <= "0000000000000000";
89         clr <= '1';
90         wait for 10 ns;
91         clr <= '0';
92         for i in 0 to 32 loop
93             wait for 10 ns;
94             x <= std_logic_vector(unsigned(x) + 1);
95         end loop;
96
97         wait;
98     end process;
99
100 END;
101
102
103
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    23:35:12 03/22/2011
6  -- Design Name:
7  -- Module Name:    x7segb_top - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity x7segb_top is
24     Port ( clk : in  STD_LOGIC;
25           btn : in  STD_LOGIC_VECTOR (3 downto 0);
26           sw : in  STD_LOGIC_VECTOR (7 downto 0);
27           a_to_g : out STD_LOGIC_VECTOR (6 downto 0);
28           an : out STD_LOGIC_VECTOR (3 downto 0);
29           dp : out  STD_LOGIC);
30 end x7segb_top;
31
32 architecture Behavioral of x7segb_top is
33 component x7segb is
34     Port(
35         x : in  STD_LOGIC_VECTOR (15 downto 0);
36         clk : in  STD_LOGIC;
37         clr : in  STD_LOGIC;
38         a_to_g : out STD_LOGIC_VECTOR (6 downto 0);
39         an : out  STD_LOGIC_VECTOR (3 downto 0);
40         dp : out  STD_LOGIC
41     );
42 end component;
43
44 signal x : STD_LOGIC_VECTOR(15 downto 0);
45
46 begin
47     --concatentate switches and 3 buttons
48     x <= sw & btn(2 downto 0) & "01010" ; --digit 0 = A
49
50     X2 : x7segb port map
51     (   x => x,
52         clk => clk,
53         clr => btn(3),
54         a_to_g => a_to_g,
55         an => an,
56         dp => dp
```

```
57      );  
58  
59  end Behavioral;  
60  
61
```

```
1
2
3  NET "a_to_g[0]" LOC = H14;
4  NET "a_to_g[1]" LOC = J17;
5  NET "a_to_g[2]" LOC = G14;
6  NET "a_to_g[3]" LOC = D16;
7  NET "a_to_g[4]" LOC = D17;
8  NET "a_to_g[5]" LOC = F18;
9  NET "a_to_g[6]" LOC = L18;
10 NET "an[0]" LOC = F17;
11 NET "an[1]" LOC = H17;
12 NET "an[2]" LOC = C18;
13 NET "an[3]" LOC = F15;
14 NET "dp" LOC = C17;
15 NET "clk" LOC = B8;
16 NET "btn[0]" LOC = B18;
17 NET "btn[1]" LOC = D18;
18 NET "btn[3]" LOC = H13;
19 NET "btn[2]" LOC = E18;
20 NET "sw[0]" LOC = G18;
21 NET "sw[1]" LOC = H18;
22 NET "sw[2]" LOC = K18;
23 NET "sw[3]" LOC = K17;
24 NET "sw[4]" LOC = L14;
25 NET "sw[5]" LOC = L13;
26 NET "sw[6]" LOC = N17;
27 NET "sw[7]" LOC = R17;
28
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    22:12:22 04/08/2011
6  -- Design Name:
7  -- Module Name:    F:/Senior Project/x7segb_top/x7segb_top_TB.vhd
8  -- Project Name:   x7segb_top
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: x7segb_top
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27 -----
28 LIBRARY ieee;
29 USE ieee.std_logic_1164.ALL;
30 USE ieee.numeric_std.ALL;
31
32 ENTITY x7segb_top_TB IS
33 END x7segb_top_TB;
34
35 ARCHITECTURE behavior OF x7segb_top_TB IS
36
37     -- Component Declaration for the Unit Under Test (UUT)
38
39     COMPONENT x7segb_top
40     PORT(
41         clk : IN  std_logic;
42         btn : IN  std_logic_vector(3 downto 0);
43         sw  : IN  std_logic_vector(7 downto 0);
44         a_to_g : OUT std_logic_vector(6 downto 0);
45         an : OUT  std_logic_vector(3 downto 0);
46         dp : OUT  std_logic
47     );
48     END COMPONENT;
49
50
51     --Inputs
52     signal clk : std_logic := '0';
53     signal btn : std_logic_vector(3 downto 0) := (others => '0');
54     signal sw  : std_logic_vector(7 downto 0) := (others => '0');
55
56     --Outputs
```



```
57     signal a_to_g : std_logic_vector(6 downto 0);
58     signal an : std_logic_vector(3 downto 0);
59     signal dp : std_logic;
60
61     -- Clock period definitions
62     constant clk_period : time := 10 ns;
63
64 BEGIN
65
66     -- Instantiate the Unit Under Test (UUT)
67     uut: x7segb_top PORT MAP (
68         clk => clk,
69         btn => btn,
70         sw => sw,
71         a_to_g => a_to_g,
72         an => an,
73         dp => dp
74     );
75
76     -- Clock process definitions
77     clk_process :process
78     begin
79         clk <= '0';
80         wait for clk_period/2;
81         clk <= '1';
82         wait for clk_period/2;
83     end process;
84
85     -- Stimulus process
86     stim_proc: process
87     begin
88         sw <= "00000000";
89
90
91         btn(3) <= '1';
92         wait for 10 ns;
93         btn(3) <= '0';
94         for i in 0 to 32 loop
95             wait for 10 ns;
96             sw <= std_logic_vector(unsigned(sw) + 1);
97         end loop;
98
99         wait;
100     end process;
101
102 END;
```

```
103
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:      16:54:54 03/23/2011
6  -- Design Name:
7  -- Module Name:      sat4bit - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity sat4bit is
24     Port ( x : in  STD_LOGIC_VECTOR (5 downto 0);
25           y : out  STD_LOGIC_VECTOR (3 downto 0));
26 end sat4bit;
27
28 architecture Behavioral of sat4bit is
29     component mux24
30     port (
31         a : in STD_LOGIC_VECTOR (3 downto 0);
32         b : in STD_LOGIC_VECTOR (3 downto 0);
33         s : in STD_LOGIC;
34         y : out STD_LOGIC_VECTOR (3 downto 0)
35     );
36 end component;
37
38 signal c0 : STD_LOGIC;
39 signal c1 : STD_LOGIC;
40 signal s : STD_LOGIC;
41 signal xi : STD_LOGIC;
42
43 begin
44     U1 : mux24
45         port map ( a(0) => xi, a(1) => xi, a(2) => xi, a(3) => x(5),
46                 b(0) => x(0), b(1) => x(1), b(2) => x(2),
47                 b(3) => x(3), s => s, y => y);
48
49     c1 <= not(x(4) xor x(3));
50     xi <= not(x(5));
51     c0 <= not(x(5) xor x(4));
52     s <= c0 and c1;
53
54 end Behavioral;
55
56
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    14:01:24 03/24/2011
6  -- Design Name:
7  -- Module Name:    x7segb11 - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19  -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22 use IEEE.STD_LOGIC_UNSIGNED.ALL;
23
24 entity x7segb11 is
25     Port ( y : in  STD_LOGIC_VECTOR(3 downto 0);
26           cclk : in  STD_LOGIC;
27           clr : in  STD_LOGIC;
28           a_to_g : out STD_LOGIC_VECTOR(6 downto 0);
29           an : out STD_LOGIC_VECTOR(3 downto 0);
30           dp : out STD_LOGIC);
31 end x7segb11;
32
33 architecture Behavioral of x7segb11 is
34
35     signal msel : STD_LOGIC;
36     signal a_g0 : STD_LOGIC_VECTOR( 6 downto 0);
37     signal a_g1 : STD_LOGIC_VECTOR( 6 downto 0);
38     signal s : STD_LOGIC_VECTOR( 1 downto 0);
39     signal digit : STD_LOGIC_VECTOR( 3 downto 0);
40     signal aen : STD_LOGIC_VECTOR( 3 downto 0);
41
42     begin
43
44     dp <= '1';
45     a_g1 <= "1111110";
46     digit <= y(3 downto 0);
47
48     aen(3) <= '0';
49     aen(2) <= '0';
50     aen(1) <= y(3);
51     aen(0) <= y(3) or y(2) or y(1) or y(0);
52
53     --Quad 4-to-1 MUX: mux44
54     process(s)
55     begin
56         case s is
```

```
57     when "00" => msel <= '0';
58     when "01" => msel <= '1'; --display minus sign
59     when "10" => msel <= '0';
60     when others => msel <= '0';
61 end case;
62 end process;
63
64
65 --7 segment decoder: hex7seg
66 process(digit)
67 begin
68     case digit is
69         when X"0" => a_g0 <= "0000001"; --0
70         when X"1" => a_g0 <= "1001111"; --1
71         when X"2" => a_g0 <= "0010010"; --2
72         when X"3" => a_g0 <= "0000110"; --3
73         when X"4" => a_g0 <= "1001100"; --4
74         when X"5" => a_g0 <= "0100100"; --5
75         when X"6" => a_g0 <= "0100000"; --6
76         when X"7" => a_g0 <= "0001101"; --7
77         when X"8" => a_g0 <= "0000000"; -- -8
78         when X"9" => a_g0 <= "0000100"; -- -7
79         when X"A" => a_g0 <= "0001000"; -- -6
80         when X"B" => a_g0 <= "1100000"; -- -5
81         when X"C" => a_g0 <= "0110001"; -- -4
82         when X"D" => a_g0 <= "1000010"; -- -3
83         when X"E" => a_g0 <= "0110000"; -- -2
84         when X"F" => a_g0 <= "0110000"; -- -1
85         when others => a_g0 <= "0000001"; --0
86     end case;
87 end process;
88
89 --2-to-1 MUX
90 process(msel, a_g1, a_g0)
91 begin
92     if msel = '1' then
93         a_to_g <= a_g1;
94     else
95         a_to_g <= a_g0;
96     end if;
97 end process;
98
99 --Digit select : ancode
100 process(s, aen)
101 begin
102     an <= "1111";
103     if aen(conv_integer(s)) = '1' then
104         an(conv_integer(s)) <= '0';
105     end if;
106 end process;
107
108 --2-bit counter
109 process(cclk, clr)
110 begin
111     if clr = '1' then
112         s <= "00";
```

```
113     elsif cclk'event and cclk = '1' then
114         s <= s + "01";
115     end if;
116 end process;
117
118 end Behavioral;
119
120
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    17:06:09 03/24/2011
6  -- Design Name:
7  -- Module Name:    sat4bit_top - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19  -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity sat4bit_top is
24     Port ( mclk : in  STD_LOGIC;
25           btn  : in  STD_LOGIC_VECTOR (3 downto 3);
26           sw   : in  STD_LOGIC_VECTOR (5 downto 0);
27           dp   : out STD_LOGIC;
28           a_to_g : out STD_LOGIC_VECTOR (6 downto 0);
29           an   : out STD_LOGIC_VECTOR (3 downto 0);
30           ld   : out STD_LOGIC_VECTOR (5 downto 0)
31     );
32 end sat4bit_top;
33
34 architecture Behavioral of sat4bit_top is
35
36     component clkdiv
37     port(
38         clr : in STD_LOGIC;
39         mclk : in STD_LOGIC;
40         clk190 : out STD_LOGIC
41     );
42 end component;
43
44     component sat4bit
45     port(
46         x : in STD_LOGIC_VECTOR(5 downto 0);
47         y : out STD_LOGIC_VECTOR(3 downto 0)
48     );
49 end component;
50
51     component x7segb11
52     port(
53         cclk : in STD_LOGIC;
54         clr : in STD_LOGIC;
55         y : in STD_LOGIC_VECTOR( 3 downto 0);
56         a_to_g : out STD_LOGIC_VECTOR(6 downto 0);
```

```
57     an : out STD_LOGIC_VECTOR(3 downto 0);
58     dp : out STD_LOGIC
59 );
60 end component;
61
62 signal clk190 : STD_LOGIC;
63 signal y : STD_LOGIC_VECTOR(3 downto 0);
64
65 begin
66   U1 : sat4bit
67     port map(
68       x => sw,
69       y => y
70     );
71
72   U2 : x7segb11
73     port map(
74       a_to_g => a_to_g,
75       an => an,
76       cclk => clk190,
77       clr => btn(3),
78       dp => dp,
79       y => y
80     );
81
82   U3 : clkdiv
83     port map(
84       clk190 => clk190,
85       clr => btn(3),
86       mclk => mclk
87     );
88
89   ld <= sw;
90
91 end Behavioral;
92
93
```

```
1
2 # PlanAhead Generated physical constraints
3
4 NET "a_to_g[0]" LOC = H14;
5 NET "a_to_g[1]" LOC = J17;
6 NET "a_to_g[2]" LOC = G14;
7 NET "a_to_g[3]" LOC = D16;
8 NET "a_to_g[4]" LOC = D17;
9 NET "a_to_g[5]" LOC = F18;
10 NET "a_to_g[6]" LOC = L18;
11 NET "an[0]" LOC = F17;
12 NET "an[1]" LOC = H17;
13 NET "an[2]" LOC = C18;
14 NET "an[3]" LOC = F15;
15 NET "btn[3]" LOC = H13;
16 NET "dp" LOC = C17;
17 NET "ld[0]" LOC = J14;
18 NET "ld[1]" LOC = J15;
19 NET "ld[2]" LOC = K15;
20 NET "ld[3]" LOC = K14;
21 NET "ld[4]" LOC = E17;
22 NET "ld[5]" LOC = P15;
23 NET "mclk" LOC = B8;
24 NET "sw[0]" LOC = G18;
25 NET "sw[1]" LOC = H18;
26 NET "sw[2]" LOC = K18;
27 NET "sw[3]" LOC = K17;
28 NET "sw[4]" LOC = L14;
29 NET "sw[5]" LOC = L13;
30
```



```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    20:24:27 03/29/2011
6  -- Design Name:
7  -- Module Name:    halfadder - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19  -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22 use IEEE.STD_LOGIC_UNSIGNED.ALL;
23
24
25 entity halfadder is
26     Port ( a : in  STD_LOGIC;
27           b : in  STD_LOGIC;
28           c : out STD_LOGIC;
29           s : out STD_LOGIC);
30 end halfadder;
31
32 architecture Behavioral of halfadder is
33
34 begin
35     s <= a xor b;
36     c <= a and b;
37
38 end Behavioral;
39
40
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:   20:26:00 03/29/2011
6  -- Design Name:
7  -- Module Name:   F:/Senior Project/halfadder/halfadder_TB.vhd
8  -- Project Name:  halfadder
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: halfadder
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30  USE ieee.numeric_std.ALL;
31
32  -- Uncomment the following library declaration if using
33  -- arithmetic functions with Signed or Unsigned values
34  --USE ieee.numeric_std.ALL;
35
36  ENTITY halfadder_TB IS
37  END halfadder_TB;
38
39  ARCHITECTURE behavior OF halfadder_TB IS
40
41      -- Component Declaration for the Unit Under Test (UUT)
42
43      COMPONENT halfadder
44      PORT(
45          a : IN  std_logic;
46          b : IN  std_logic;
47          c : OUT std_logic;
48          s : OUT std_logic
49      );
50      END COMPONENT;
51
52
53      --Inputs
54      signal a : std_logic := '0';
55      signal b : std_logic := '0';
56
```

```
57  --Outputs
58  signal c : std_logic;
59  signal s : std_logic;
60  -- No clocks detected in port list. Replace <clock> below with
61  -- appropriate port name
62
63
64  BEGIN
65
66  -- Instantiate the Unit Under Test (UUT)
67  uut: halfadder PORT MAP (
68      a => a,
69      b => b,
70      c => c,
71      s => s
72  );
73
74
75
76  -- Stimulus process
77  stim_proc: process
78  begin
79      a <= '0';
80      b <= '0';
81
82      wait for 10 ns;
83
84      a <= '0';
85      b <= '1';
86
87      wait for 10 ns;
88
89      a <= '1';
90      b <= '0';
91
92      wait for 10 ns;
93
94      a <= '1';
95      b <= '1';
96      wait for 10 ns;
97
98
99      wait;
100  end process;
101
102  END;
103
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    20:35:23 03/29/2011
6  -- Design Name:
7  -- Module Name:    fulladder - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23
24 entity fulladder is
25     Port ( a : in  STD_LOGIC;
26           b : in  STD_LOGIC;
27           cin : in  STD_LOGIC;
28           cout : out STD_LOGIC;
29           s : out  STD_LOGIC);
30 end fulladder;
31
32 architecture Behavioral of fulladder is
33     component halfadder
34     port(
35         a : in STD_LOGIC;
36         b : in STD_LOGIC;
37         c : out STD_LOGIC;
38         s : out STD_LOGIC
39     );
40 end component;
41
42 signal c1 : STD_LOGIC;
43 signal c2 : STD_LOGIC;
44 signal s1 : STD_LOGIC;
45
46 begin
47     U1 : halfadder
48         port map(
49             a => a, b => b, c => c1, s => s1);
50
51     U2: halfadder
52         port map(
53             a => s1, b => cin, c => c2, s => s );
54
55     cout <= c2 or c1;
56
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    20:43:45 03/29/2011
6  -- Design Name:
7  -- Module Name:    F:/Senior Project/fulladder/fulladde_TB.vhd
8  -- Project Name:   fulladder
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: fulladder
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30
31
32  ENTITY fulladde_TB IS
33  END fulladde_TB;
34
35  ARCHITECTURE behavior OF fulladde_TB IS
36
37      -- Component Declaration for the Unit Under Test (UUT)
38
39      COMPONENT fulladder
40      PORT(
41          a : IN  std_logic;
42          b : IN  std_logic;
43          cin : IN  std_logic;
44          cout : OUT std_logic;
45          s : OUT std_logic
46      );
47      END COMPONENT;
48
49
50      --Inputs
51      signal a : std_logic := '0';
52      signal b : std_logic := '0';
53      signal cin : std_logic := '0';
54
55      --Outputs
56      signal cout : std_logic;
```

```
57  end Behavioral;  
58  
59
```

```
57     signal s : std_logic;
58     -- No clocks detected in port list. Replace <clock> below with
59     -- appropriate port name
60
61 BEGIN
62
63     -- Instantiate the Unit Under Test (UUT)
64     uut: fulladder PORT MAP (
65         a => a,
66         b => b,
67         cin => cin,
68         cout => cout,
69         s => s
70     );
71
72
73
74
75     -- Stimulus process
76     stim_proc: process
77     begin
78
79
80         cin <= '0';
81         a <= '0';
82         b <= '0';
83         wait for 10 ns;
84         cin <= '0';
85         a <= '0';
86         b <= '1';
87         wait for 10 ns;
88         cin <= '0';
89         a <= '1';
90         b <= '0';
91         wait for 10 ns;
92         cin <= '0';
93         a <= '1';
94         b <= '1';
95         wait for 10 ns;
96         cin <= '1';
97         a <= '0';
98         b <= '0';
99         wait for 10 ns;
100        cin <= '1';
101        a <= '0';
102        b <= '1';
103        wait for 10 ns;
104        cin <= '1';
105        a <= '1';
106        b <= '0';
107        wait for 10 ns;
108        cin <= '1';
109        a <= '1';
110        b <= '1';
111
112        wait;
```

```
113     end process;  
114  
115     END;  
116
```



```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:      10:33:05 03/31/2011
6  -- Design Name:
7  -- Module Name:      adder4 - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity adder4 is
24     Port ( cin : in  STD_LOGIC;
25           a  : in  STD_LOGIC_VECTOR (3 downto 0);
26           b  : in  STD_LOGIC_VECTOR (3 downto 0);
27           cout : out STD_LOGIC;
28           s  : out  STD_LOGIC_VECTOR (3 downto 0));
29 end adder4;
30
31 architecture Behavioral of adder4 is
32     component fulladder
33     port(
34         a : in  STD_LOGIC;
35         b : in  STD_LOGIC;
36         cin : in  STD_LOGIC;
37         cout : out STD_LOGIC;
38         s : out  STD_LOGIC
39     );
40 end component;
41
42 signal c1 : STD_LOGIC;
43 signal c2 : STD_LOGIC;
44 signal c3 : STD_LOGIC;
45
46 begin
47     U1 : fulladder
48     port map(
49         a => a(2), b => b(2), cin => c2, cout => c3,
50         s => s(2));
51
52     U2 : fulladder
53     port map(
54         a => a(3), b => b(3), cin => c3, cout => cout,
55         s => s(3));
56
```

```
57  U3 : fulladder
58      port map(
59          a => a(1), b => b(1), cin => c1, cout => c2,
60          s => s(1));
61
62  U4 : fulladder
63      port map(
64          a => a(0), b => b(0), cin => cin, cout => c1,
65          s => s(0));
66  end Behavioral;
67
68
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:   10:48:26 03/31/2011
6  -- Design Name:
7  -- Module Name:   H:/Senior Project/adder4/adder4_TB.vhd
8  -- Project Name:  adder4
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: adder4
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30  USE ieee.numeric_std.ALL;
31
32
33  ENTITY adder4_TB IS
34  END adder4_TB;
35
36  ARCHITECTURE behavior OF adder4_TB IS
37
38      -- Component Declaration for the Unit Under Test (UUT)
39
40      COMPONENT adder4
41      PORT(
42          cin : IN  std_logic;
43          a : IN  std_logic_vector(3 downto 0);
44          b : IN  std_logic_vector(3 downto 0);
45          cout : OUT std_logic;
46          s : OUT  std_logic_vector(3 downto 0)
47      );
48      END COMPONENT;
49
50
51      --Inputs
52      signal cin : std_logic := '0';
53      signal a : std_logic_vector(3 downto 0) := (others => '0');
54      signal b : std_logic_vector(3 downto 0) := (others => '0');
55
56      --Outputs
```

```
57     signal cout : std_logic;
58     signal s : std_logic_vector(3 downto 0);
59     -- No clocks detected in port list. Replace <clock> below with
60     -- appropriate port name
61
62
63
64 BEGIN
65
66     -- Instantiate the Unit Under Test (UUT)
67     uut: adder4 PORT MAP (
68         cin => cin,
69         a => a,
70         b => b,
71         cout => cout,
72         s => s
73     );
74
75
76
77     -- Stimulus process
78     stim_proc: process
79     begin
80         a <= "0000";
81         b <= "1011";
82         cin <= '0';
83
84
85         for i in 0 to 16 loop
86             wait for 20 ns;
87             a <= std_logic_vector( unsigned(a)+1);
88         end loop;
89
90
91         wait;
92     end process;
93
94 END;
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    14:24:56 04/14/2011
6  -- Design Name:
7  -- Module Name:    adder4_top - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19  -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity adder4_top is
24     Port ( mclk : in  STD_LOGIC;
25           btn  : in  STD_LOGIC_VECTOR (3 downto 3);
26           sw   : in  STD_LOGIC_VECTOR (7 downto 0);
27           a_to_g : out STD_LOGIC_VECTOR (6 downto 0);
28           an   : out STD_LOGIC_VECTOR (3 downto 0);
29           dp   : out STD_LOGIC;
30           ld   : out STD_LOGIC_VECTOR (7 downto 0));
31 end adder4_top;
32
33 architecture Behavioral of adder4_top is
34 component adder4 is
35     Port ( cin : in  STD_LOGIC;
36           a  : in  STD_LOGIC_VECTOR (3 downto 0);
37           b  : in  STD_LOGIC_VECTOR (3 downto 0);
38           cout : out STD_LOGIC;
39           s  : out STD_LOGIC_VECTOR (3 downto 0));
40 end component;
41
42 component x7segb is
43     Port ( x : in  STD_LOGIC_VECTOR (15 downto 0);
44           clk : in  STD_LOGIC;
45           clr : in  STD_LOGIC;
46           a_to_g : out STD_LOGIC_VECTOR (6 downto 0);
47           an   : out STD_LOGIC_VECTOR (3 downto 0);
48           dp   : out STD_LOGIC);
49 end component;
50
51 component clkdiv is
52     Port ( mclk : in  STD_LOGIC;
53           clr : in  STD_LOGIC;
54           clk190 : out STD_LOGIC
55           -- clk48 : out  STD_LOGIC
56           --clk3 : out STD_LOGIC
```

```
57         );
58     end component;
59
60     signal clk190, clr, c4, cin : STD_LOGIC;
61     signal x : STD_LOGIC_VECTOR (15 downto 0);
62     signal sum : STD_LOGIC_VECTOR (3 downto 0);
63
64     begin
65
66         cin <= '0';
67         x <= sw & "000" & c4 & sum;
68         clr <= btn(3);
69         ld <= sw;
70
71         U1 : adder4
72             port map( cin => cin, a => sw(7 downto 4), b => sw(3 downto 0), cout => c4, s =>
sum);
73
74         U2 : x7segb
75             port map( x => x, clk => clk190, clr => clr, a_to_g => a_to_g, an => an, dp => dp
);
76
77         U3 : clkdiv
78             port map( mclk => mclk, clr => clr, clk190 => clk190);
79
80     end Behavioral;
81
82
```

```
1  NET "ld[0]" LOC = J14;
2  NET "ld[1]" LOC = J15;
3  NET "ld[2]" LOC = K15;
4  NET "ld[3]" LOC = K14;
5  NET "ld[4]" LOC = E17;
6  NET "ld[5]" LOC = P15;
7  NET "ld[6]" LOC = F4;
8  NET "ld[7]" LOC = R4;
9  NET "sw[0]" LOC = G18;
10 NET "sw[1]" LOC = H18;
11 NET "sw[2]" LOC = K18;
12 NET "sw[3]" LOC = K17;
13 NET "sw[4]" LOC = L14;
14 NET "sw[5]" LOC = L13;
15 NET "sw[6]" LOC = N17;
16 NET "sw[7]" LOC = R17;
17 NET "a_to_g[0]" LOC = H14;
18 NET "a_to_g[1]" LOC = J17;
19 NET "a_to_g[2]" LOC = G14;
20 NET "a_to_g[3]" LOC = D16;
21 NET "a_to_g[4]" LOC = D17;
22 NET "a_to_g[5]" LOC = F18;
23 NET "a_to_g[6]" LOC = L18;
24 NET "an[0]" LOC = F17;
25 NET "an[1]" LOC = H17;
26 NET "an[2]" LOC = C18;
27 NET "an[3]" LOC = F15;
28 NET "dp" LOC = C17;
29 NET "mclk" LOC = B8;
30 NET "btn[3]" LOC = H13;
31
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    18:31:39 04/01/2011
6  -- Design Name:
7  -- Module Name:    adder4b - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19  -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22 use IEEE.STD_LOGIC_UNSIGNED.ALL;
23
24 entity adder4b is
25     Port ( a : in  STD_LOGIC_VECTOR (3 downto 0);
26           b : in  STD_LOGIC_VECTOR (3 downto 0);
27           s : out STD_LOGIC_VECTOR (3 downto 0);
28           cf : out STD_LOGIC);
29 end adder4b;
30
31 architecture Behavioral of adder4b is
32
33 begin
34 process(a,b)
35     variable temp : STD_LOGIC_VECTOR(4 downto 0);
36 begin
37     temp := ('0' & a) + ('0' & b);
38     s <= temp(3 downto 0);
39     cf <= temp(4);
40     s <= a + b;
41 end process;
42
43 end Behavioral;
44
45
```



```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:   18:34:38 04/01/2011
6  -- Design Name:
7  -- Module Name:   F:/Senior Project/adder4b/adder4b_TB.vhd
8  -- Project Name:  adder4b
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: adder4b
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30  USE ieee.numeric_std.ALL;
31
32
33  ENTITY adder4b_TB IS
34  END adder4b_TB;
35
36  ARCHITECTURE behavior OF adder4b_TB IS
37
38      -- Component Declaration for the Unit Under Test (UUT)
39
40      COMPONENT adder4b
41      PORT (
42          a : IN  std_logic_vector(3 downto 0);
43          b : IN  std_logic_vector(3 downto 0);
44          s : OUT std_logic_vector(3 downto 0);
45          cf : OUT std_logic
46      );
47      END COMPONENT;
48
49
50      --Inputs
51      signal a : std_logic_vector(3 downto 0) := (others => '0');
52      signal b : std_logic_vector(3 downto 0) := (others => '0');
53
54      --Outputs
55      signal s : std_logic_vector(3 downto 0);
56      signal cf : std_logic;
```

```
57      -- No clocks detected in port list. Replace <clock> below with
58      -- appropriate port name
59
60
61  BEGIN
62
63      -- Instantiate the Unit Under Test (UUT)
64      uut: adder4b PORT MAP (
65          a => a,
66          b => b,
67          s => s,
68          cf => cf
69      );
70
71
72      -- Stimulus process
73      stim_proc: process
74      begin
75          a <= "0000";
76          b <= "1011";
77
78
79
80          for i in 0 to 16 loop
81              wait for 20 ns;
82              a <= std_logic_vector(unsigned(a)+1);
83          end loop;
84
85
86          wait;
87      end process;
88
89  END;
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    18:43:41 04/01/2011
6  -- Design Name:
7  -- Module Name:    adder - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22 use IEEE.STD_LOGIC_UNSIGNED.ALL;
23
24
25 entity adder is
26     generic( N: integer :=8);
27     Port ( a : in  STD_LOGIC_VECTOR (N-1 downto 0);
28           b : in  STD_LOGIC_VECTOR (N-1 downto 0);
29           y : out STD_LOGIC_VECTOR (N-1 downto 0));
30 end adder;
31
32 architecture Behavioral of adder is
33
34 begin
35     process(a, b)
36     begin
37         y<= a + b;
38     end process;
39
40 end Behavioral;
41
42
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:   18:46:09 04/01/2011
6  -- Design Name:
7  -- Module Name:   F:/Senior Project/adder/adder_TB.vhd
8  -- Project Name:  adder
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: adder
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30  USE ieee.numeric_std.ALL;
31
32  ENTITY adder_TB IS
33  END adder_TB;
34
35  ARCHITECTURE behavior OF adder_TB IS
36
37      -- Component Declaration for the Unit Under Test (UUT)
38
39      COMPONENT adder
40      PORT(
41          a : IN  std_logic_vector(7 downto 0);
42          b : IN  std_logic_vector(7 downto 0);
43          y : OUT std_logic_vector(7 downto 0)
44      );
45      END COMPONENT;
46
47
48      --Inputs
49      signal a : std_logic_vector(7 downto 0) := (others => '0');
50      signal b : std_logic_vector(7 downto 0) := (others => '0');
51
52      --Outputs
53      signal y : std_logic_vector(7 downto 0);
54      -- No clocks detected in port list. Replace <clock> below with
55      -- appropriate port name
56
```

```
57
58 BEGIN
59
60     -- Instantiate the Unit Under Test (UUT)
61     uut: adder PORT MAP (
62         a => a,
63         b => b,
64         y => y
65     );
66
67
68     -- Stimulus process
69     stim_proc: process
70     begin
71         a <= "00010001";
72         b <= "00110011";
73
74         for i in 0 to 16 loop
75             wait for 20 ns;
76             a <= std_logic_vector(unsigned(a) + X"22");
77         end loop;
78
79         wait;
80     end process;
81
82 END;
83
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    13:58:08 04/14/2011
6  -- Design Name:
7  -- Module Name:    adder_top - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19  -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity adder_top is
24     Port ( mclk : in  STD_LOGIC;
25           btn : in  STD_LOGIC_VECTOR (3 downto 3);
26           sw : in  STD_LOGIC_VECTOR (7 downto 0);
27           a_to_g : out STD_LOGIC_VECTOR (6 downto 0);
28           an : out  STD_LOGIC_VECTOR (3 downto 0);
29           dp : out  STD_LOGIC;
30           ld : out  STD_LOGIC_VECTOR (7 downto 0));
31 end adder_top;
32
33 architecture Behavioral of adder_top is
34 component clkdiv is
35     Port ( mclk : in  STD_LOGIC;
36           clr : in  STD_LOGIC;
37           clk190 : out STD_LOGIC
38           );
39 end component;
40
41
42 component adder is
43     generic( N: integer := 4);
44     Port ( a : in  STD_LOGIC_VECTOR (N-1 downto 0);
45           b : in  STD_LOGIC_VECTOR (N-1 downto 0);
46           y : out STD_LOGIC_VECTOR (N-1 downto 0));
47 end component;
48
49 component x7segb is
50     Port ( x : in  STD_LOGIC_VECTOR (15 downto 0);
51           clk : in  STD_LOGIC;
52           clr : in  STD_LOGIC;
53           a_to_g : out STD_LOGIC_VECTOR (6 downto 0);
54           an : out  STD_LOGIC_VECTOR (3 downto 0);
55           dp : out  STD_LOGIC);
56 end component;
```

```
57
58
59 signal clk190, clr : STD_LOGIC;
60 signal g : STD_LOGIC_VECTOR (11 downto 0);
61 signal s : STD_LOGIC_VECTOR (3 downto 0);
62 signal x : STD_LOGIC_VECTOR (15 downto 0);
63
64 begin
65
66   clr <= btn(3);
67   g <= "0000000000000";
68
69
70
71   U1 : clkdiv
72     port map( mclk => mclk, clr => clr, clk190 => clk190);
73
74   U2 : adder
75     generic map( N => 4)
76     port map( a => sw(7 downto 4), b => sw(3 downto 0), y => s(3 downto 0));
77
78   x <= g & s;
79
80   U3 : x7segb
81     port map( clk => clk190, clr => clr, x => x, a_to_g => a_to_g, an => an, dp => dp
82     );
83   ld <= sw;
84   end Behavioral;
85
86
```

```
1  NET "ld[0]" LOC = J14;
2  NET "ld[1]" LOC = J15;
3  NET "ld[2]" LOC = K15;
4  NET "ld[3]" LOC = K14;
5  NET "ld[4]" LOC = E17;
6  NET "ld[5]" LOC = P15;
7  NET "ld[6]" LOC = F4;
8  NET "ld[7]" LOC = R4;
9  NET "sw[0]" LOC = G18;
10 NET "sw[1]" LOC = H18;
11 NET "sw[2]" LOC = K18;
12 NET "sw[3]" LOC = K17;
13 NET "sw[4]" LOC = L14;
14 NET "sw[5]" LOC = L13;
15 NET "sw[6]" LOC = N17;
16 NET "sw[7]" LOC = R17;
17 NET "a_to_g[0]" LOC = H14;
18 NET "a_to_g[1]" LOC = J17;
19 NET "a_to_g[2]" LOC = G14;
20 NET "a_to_g[3]" LOC = D16;
21 NET "a_to_g[4]" LOC = D17;
22 NET "a_to_g[5]" LOC = F18;
23 NET "a_to_g[6]" LOC = L18;
24 NET "an[0]" LOC = F17;
25 NET "an[1]" LOC = H17;
26 NET "an[2]" LOC = C18;
27 NET "an[3]" LOC = F15;
28 NET "dp" LOC = C17;
29 NET "mclk" LOC = B8;
30 NET "btn[3]" LOC = H13;
31
```



```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    22:46:44 04/01/2011
6  -- Design Name:
7  -- Module Name:    comp - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19  -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity comp is
24     generic( N:integer := 8);
25     Port ( x : in  STD_LOGIC_VECTOR (N-1 downto 0);
26           y : in  STD_LOGIC_VECTOR (N-1 downto 0);
27           gt : out STD_LOGIC;
28           eq : out STD_LOGIC;
29           lt : out STD_LOGIC);
30 end comp;
31
32 architecture Behavioral of comp is
33
34 begin
35 process(x , y)
36 begin
37     gt <= '0';
38     eq <= '0';
39     lt <= '0';
40     if(x > y) then
41         gt <= '1';
42     elsif (x = y) then
43         eq <= '1';
44     elsif (x < y) then
45         lt <= '1';
46     end if;
47 end process;
48 end Behavioral;
49
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    22:48:39 04/01/2011
6  -- Design Name:
7  -- Module Name:    F:/Senior Project/comp/comp_TB.vhd
8  -- Project Name:   comp
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: comp
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30  USE ieee.numeric_std.ALL;
31
32  ENTITY comp_TB IS
33  END comp_TB;
34
35  ARCHITECTURE behavior OF comp_TB IS
36
37      -- Component Declaration for the Unit Under Test (UUT)
38
39      COMPONENT comp
40      PORT(
41          x : IN  std_logic_vector(7 downto 0);
42          y : IN  std_logic_vector(7 downto 0);
43          gt : OUT std_logic;
44          eq : OUT std_logic;
45          lt : OUT std_logic
46      );
47      END COMPONENT;
48
49
50      --Inputs
51      signal x : std_logic_vector(7 downto 0) := (others => '0');
52      signal y : std_logic_vector(7 downto 0) := (others => '0');
53
54      --Outputs
55      signal gt : std_logic;
56      signal eq : std_logic;
```

```
57     signal lt : std_logic;
58
59
60 BEGIN
61
62     -- Instantiate the Unit Under Test (UUT)
63     uut: comp PORT MAP (
64         x => x,
65         y => y,
66         gt => gt,
67         eq => eq,
68         lt => lt
69     );
70
71
72
73     -- Stimulus process
74     stim_proc: process
75     begin
76         x <= X"15";
77         y <= X"75";
78
79         for i in 0 to 16 loop
80             wait for 25 ns;
81             x <= std_logic_vector(unsigned(x) + X"10");
82         end loop;
83
84
85
86         wait;
87     end process;
88
89 END;
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    21:48:37 04/01/2011
6  -- Design Name:
7  -- Module Name:    comp_top - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity comp_top is
24     Port ( sw : in  STD_LOGIC_VECTOR (7 downto 0);
25           ld : out STD_LOGIC_VECTOR (4 downto 2));
26 end comp_top;
27
28 architecture Behavioral of comp_top is
29
30     component comp is
31         generic( N:integer := 8);
32         Port ( x : in  STD_LOGIC_VECTOR (N-1 downto 0);
33               y : in  STD_LOGIC_VECTOR (N-1 downto 0);
34               gt : out STD_LOGIC;
35               eq : out STD_LOGIC;
36               lt : out STD_LOGIC
37         );
38     end component;
39
40     begin
41
42
43     U1 : comp
44     generic map (N => 4)
45     port map(
46         x => sw(7 downto 4),
47         y => sw(3 downto 0),
48         eq => ld(3),
49         gt => ld(4),
50         lt => ld(2)
51     );
52
53 end Behavioral;
54
55
```

```
1
2  # PlanAhead Generated phswsical constraints
3
4  NET "ld[2]" LOC = K15;
5  NET "ld[3]" LOC = K14;
6  NET "ld[4]" LOC = E17;
7  NET "sw[0]" LOC = G18;
8  NET "sw[1]" LOC = H18;
9  NET "sw[2]" LOC = K18;
10 NET "sw[3]" LOC = K17;
11 NET "sw[4]" LOC = L14;
12 NET "sw[5]" LOC = L13;
13 NET "sw[6]" LOC = N17;
14 NET "sw[7]" LOC = R17;
15
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    17:13:07 04/10/2011
6  -- Design Name:
7  -- Module Name:    edgeDFF - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23
24 entity edgeDFF is
25     Port ( set : in  STD_LOGIC;
26           clk  : in  STD_LOGIC;
27           clr  : in  STD_LOGIC;
28           D    : in  STD_LOGIC;
29           q    : out STD_LOGIC;
30           notq : out STD_LOGIC);
31 end edgeDFF;
32
33 architecture Behavioral of edgeDFF is
34
35     signal f1, f2, f3, f4, f5, f6 : STD_LOGIC;
36
37     begin
38
39
40     f1 <= not( not(set) and f2 and f4);
41     f2 <= not( not clr and f1 and f5);
42     f3 <= not(not(set) and f6 and f4);
43     f4 <= not(f3 and clk and not(clr));
44     f5 <= not( f4 and not(set) and clk and f6);
45     f6 <= not (f5 and D and not(clr));
46
47
48     q <= f1;
49     notq <= f2;
50
51     end Behavioral;
52
53
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:   17:01:30 04/15/2011
6  -- Design Name:
7  -- Module Name:   F:/Senior Project/Example 16/edgeDFF/edgeDFF_TB.vhd
8  -- Project Name:  edgeDFF
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: edgeDFF
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30
31  -- Uncomment the following library declaration if using
32  -- arithmetic functions with Signed or Unsigned values
33  --USE ieee.numeric_std.ALL;
34
35  ENTITY edgeDFF_TB IS
36  END edgeDFF_TB;
37
38  ARCHITECTURE behavior OF edgeDFF_TB IS
39
40      -- Component Declaration for the Unit Under Test (UUT)
41
42      COMPONENT edgeDFF
43      PORT(
44          clk : IN  std_logic;
45          clr : IN  std_logic;
46          set : IN  std_logic;
47          D   : IN  std_logic;
48          q   : OUT std_logic;
49          notq : OUT std_logic
50      );
51      END COMPONENT;
52
53
54      --Inputs
55      signal clk : std_logic := '0';
56      signal clr : std_logic := '0';
```

```
57     signal set : std_logic := '0';
58     signal D : std_logic := '0';
59
60     --Outputs
61     signal q : std_logic;
62     signal notq : std_logic;
63
64     -- Clock period definitions
65     constant clk_period : time := 10 ns;
66     constant d_period : time := 40 ns;
67
68 BEGIN
69
70     -- Instantiate the Unit Under Test (UUT)
71     uut: edgeDFF PORT MAP (
72         clk => clk,
73         clr => clr,
74         set => set,
75         D => D,
76         q => q,
77         notq => notq
78     );
79
80     -- Clock process definitions
81     clk_process :process
82     begin
83         clk <= '0';
84         wait for clk_period/2;
85         clk <= '1';
86         wait for clk_period/2;
87     end process;
88
89
90     d_process :process
91     begin
92         D <= '0';
93         wait for d_period;
94         D <= '1';
95         wait for d_period;
96     end process;
97
98     -- Stimulus process
99     stim_proc: process
100     begin
101         clr <= '1';
102         set <= '0';
103         wait for 10 ns;
104         clr <= '0';
105         wait for 10 ns;
106
107         set <= '1';
108         wait for 10 ns;
109         set <= '0';
110         wait for 100 ns;
111
112
```



```
113         wait;  
114     end process;  
115  
116     END;  
117
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    23:07:40 04/01/2011
6  -- Design Name:
7  -- Module Name:    dff - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19  -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23 entity dff is
24     Port ( clr : in  STD_LOGIC;
25           clk : in  STD_LOGIC;
26           D : in  STD_LOGIC;
27           q : out STD_LOGIC
28           );
29 end dff;
30
31 architecture Behavioral of dff is
32
33 begin
34 process(clk, clr)
35 begin
36     if(clr = '1') then
37         q <= '0';
38     elsif(rising_edge(clk)) then
39         q <= D;
40     end if;
41 end process;
42
43 end Behavioral;
44
45
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:   19:19:08 04/02/2011
6  -- Design Name:
7  -- Module Name:   F:/Senior Project/dff/dff_TB.vhd
8  -- Project Name:  dff
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: dff
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30
31  -- Uncomment the following library declaration if using
32  -- arithmetic functions with Signed or Unsigned values
33  --USE ieee.numeric_std.ALL;
34
35  ENTITY dff_TB IS
36  END dff_TB;
37
38  ARCHITECTURE behavior OF dff_TB IS
39
40      -- Component Declaration for the Unit Under Test (UUT)
41
42      COMPONENT dff
43      PORT(
44          clk : IN  std_logic;
45          clr : IN  std_logic;
46          D : IN  std_logic;
47          q : OUT std_logic
48      );
49      END COMPONENT;
50
51
52      --Inputs
53      signal clk : std_logic := '0';
54      signal clr : std_logic := '0';
55      signal D : std_logic := '0';
56
```

```
57  --Outputs
58  signal q : std_logic;
59
60  -- Clock period definitions
61  constant clk_period : time := 10 ns;
62  constant d_period : time := 100 ns;
63
64  BEGIN
65
66  -- Instantiate the Unit Under Test (UUT)
67  uut: dff PORT MAP (
68      clk => clk,
69      clr => clr,
70      D => D,
71      q => q
72  );
73
74  -- Clock process definitions
75  clk_process :process
76  begin
77      clk <= '0';
78      wait for clk_period/2;
79      clk <= '1';
80      wait for clk_period/2;
81  end process;
82
83  d_process :process
84  begin
85      D <= '0';
86      wait for D_period;
87      D <= '1';
88      wait for D_period;
89  end process;
90
91  -- Stimulus process
92  stim_proc: process
93  begin
94      for i in 0 to 1 loop
95          clr <= '1';
96          wait for 10 ns;
97          clr <= '0';
98          wait for 100 ns;
99      end loop;
100
101      wait;
102  end process;
103
104  END;
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    19:30:44 04/02/2011
6  -- Design Name:
7  -- Module Name:    divideby2counter - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22
23
24 entity divideby2counter is
25     Port ( btn : in  STD_LOGIC_VECTOR (3 downto 3);
26           mclk : in  STD_LOGIC;
27           ld : out STD_LOGIC_VECTOR (3 downto 2));
28 end divideby2counter;
29
30 architecture Behavioral of divideby2counter is
31 component dff is
32     Port ( clr : in  STD_LOGIC;
33           clk : in  STD_LOGIC;
34           D : in  STD_LOGIC;
35           q : out STD_LOGIC
36           );
37 end component;
38
39 component clkdiv is
40     Port ( mclk : in  STD_LOGIC;
41           clr : in  STD_LOGIC;
42           clk3 : out STD_LOGIC
43           );
44 end component;
45
46 signal D, clk3, q0 : std_logic;
47 begin
48     ld(2) <= q0;
49
50     D <= not(q0);
51
52     U1: dff
53     port map (
54         clk => clk3, clr => btn(3), D => D, q => q0
55     );
56
```

```
57  U2: clkdiv
58      port map(
59          mclk => mclk, clr => btn(3), clk3 => clk3
60      );
61
62  ld(3) <= clk3;
63
64  end Behavioral;
65
66
```

```
1
2  # PlanAhead Generated physical constraints
3
4  NET "btn[3]" LOC = H13;
5  NET "ld[2]" LOC = K15;
6  NET "ld[3]" LOC = K14;
7  NET "mclk" LOC = B8;
8
```

```
57
58     -- Clock period definitions
59     constant mclk_period : time := 10 ns;
60
61 BEGIN
62
63     -- Instantiate the Unit Under Test (UUT)
64     uut: divideby2counter PORT MAP (
65         btn => btn,
66         mclk => mclk,
67         ld => ld
68     );
69
70     -- Clock process definitions
71     mclk_process :process
72     begin
73         mclk <= '0';
74         wait for mclk_period/2;
75         mclk <= '1';
76         wait for mclk_period/2;
77     end process;
78
79
80     -- Stimulus process
81     stim_proc: process
82     begin
83         btn(3) <= '1';
84         wait for 10 ns;
85         btn(3) <= '0';
86         wait for 100 ns;
87         wait;
88     end process;
89
90 END;
```



```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:   18:45:03 04/05/2011
6  -- Design Name:
7  -- Module Name:   F:/Senior Project/divideby2counter/divideby2counter_TB.vhd
8  -- Project Name:  divideby2counter
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: divideby2counter
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30
31  -- Uncomment the following library declaration if using
32  -- arithmetic functions with Signed or Unsigned values
33  --USE ieee.numeric_std.ALL;
34
35  ENTITY divideby2counter_TB IS
36  END divideby2counter_TB;
37
38  ARCHITECTURE behavior OF divideby2counter_TB IS
39
40      -- Component Declaration for the Unit Under Test (UUT)
41
42      COMPONENT divideby2counter
43      PORT(
44          btn : IN  std_logic_vector(3 downto 3);
45          mclk : IN  std_logic;
46          ld : OUT std_logic_vector(3 downto 2)
47      );
48      END COMPONENT;
49
50
51      --Inputs
52      signal btn : std_logic_vector(3 downto 3) := (others => '0');
53      signal mclk : std_logic := '0';
54
55      --Outputs
56      signal ld : std_logic_vector(3 downto 2);
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    19:44:40 04/05/2011
6  -- Design Name:
7  -- Module Name:    reglbit - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19  -----
20 library IEEE;
21 use IEEE.STD_LOGIC_1164.ALL;
22 entity reglbit is
23     Port ( clr : in  STD_LOGIC;
24           clk : in  STD_LOGIC;
25           load : in  STD_LOGIC;
26           inp : in  STD_LOGIC;
27           q : inout STD_LOGIC);
28 end reglbit;
29
30 architecture Behavioral of reglbit is
31 component dff is
32     port(
33         clr : in  STD_LOGIC;
34         clk : in  STD_LOGIC;
35         D : in  STD_LOGIC;
36         q : out  STD_LOGIC
37     );
38 end component;
39
40 signal nld, a1, a2, D : std_logic;
41
42 begin
43
44 U1: dff
45     port map(
46         D => D, clk => clk, clr => clr, q => q
47     );
48
49 nld <= not load;
50 a1 <= nld and q;
51 a2 <= load and inp;
52 D <= a1 or a2;
53
54 end Behavioral;
55
56
```

```
1  -----
2  -- Company:
3  -- Engineer:
4  --
5  -- Create Date:    18:04:50 04/12/2011
6  -- Design Name:
7  -- Module Name:    F:/Senior Project/reglbit/reglbit_TB.vhd
8  -- Project Name:   reglbit
9  -- Target Device:
10 -- Tool versions:
11 -- Description:
12 --
13 -- VHDL Test Bench Created by ISE for module: reglbit
14 --
15 -- Dependencies:
16 --
17 -- Revision:
18 -- Revision 0.01 - File Created
19 -- Additional Comments:
20 --
21 -- Notes:
22 -- This testbench has been automatically generated using types std_logic and
23 -- std_logic_vector for the ports of the unit under test.  Xilinx recommends
24 -- that these types always be used for the top-level I/O of a design in order
25 -- to guarantee that the testbench will bind correctly to the post-implementation
26 -- simulation model.
27  -----
28  LIBRARY ieee;
29  USE ieee.std_logic_1164.ALL;
30
31  ENTITY reglbit_TB IS
32  END reglbit_TB;
33
34  ARCHITECTURE behavior OF reglbit_TB IS
35
36      -- Component Declaration for the Unit Under Test (UUT)
37
38      COMPONENT reglbit
39      PORT(
40          clr : IN  std_logic;
41          clk : IN  std_logic;
42          load : IN  std_logic;
43          inp : IN  std_logic;
44          q : INOUT std_logic
45      );
46      END COMPONENT;
47
48
49      --Inputs
50      signal clk : std_logic := '0';
51      signal clr : std_logic := '0';
52      signal load : std_logic := '0';
53      signal inp : std_logic := '0';
54
55      --BiDirs
56      signal q : std_logic;
```

```
57
58     -- Clock period definitions
59     constant clk_period : time := 10 ns;
60
61 BEGIN
62
63     -- Instantiate the Unit Under Test (UUT)
64     uut: reglbit PORT MAP (
65         clr => clr,
66         clk => clk,
67         load => load,
68         inp => inp,
69         q => q
70     );
71
72     -- Clock process definitions
73     clk_process : process
74     begin
75         clk <= '0';
76         wait for clk_period/2;
77         clk <= '1';
78         wait for clk_period/2;
79     end process;
80
81     --load process
82     load_process: process
83     begin
84         load <= '0';
85         wait for 40 ns;
86         load <= '1';
87         wait for 20 ns;
88         load <= '0';
89         wait for 40 ns;
90     end process;
91
92     --input process
93     inp_process: process
94     begin
95         inp <= '1';
96         wait for 100 ns;
97         inp <= '0';
98         wait for 100ns;
99     end process;
100
101
102
103     -- Stimulus process
104     stim_proc: process
105     begin
106         clr <= '1' ;
107         wait for 10 ns;
108         clr <= '0';
109         wait for 10 ns;
110
111         wait;
112     end process;
```

113

114 END;

115